

7DT Single-band Matching in COSMOS1

From 20 band catalogs to a 75,259-object union catalog

20 band catalogs

78,882 selected detections



7DT cross-band match

19,489 7DT masters



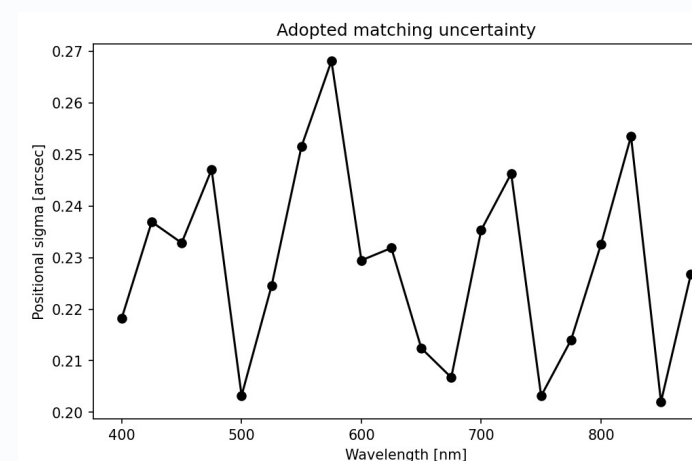
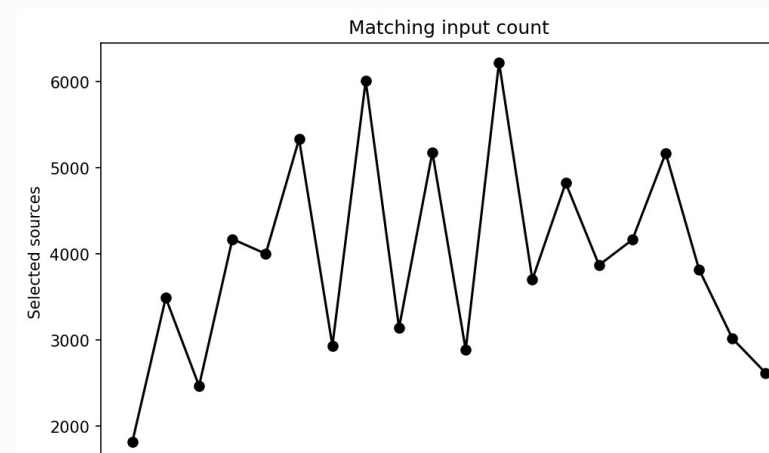
FINALCAT sky match

2,977 shared objects



Union catalog

75,259 total objects



Matching uses a position error for each band.

7DT Cross-band Matching

Use position error to judge whether detections belong to one object

01

New band source

02

3 σ candidate search

03

One-to-one match

04

Weighted master
update

Band position error

$$\sigma_b = \text{FWHM}_b / (2.355 \times \text{S/N}_{\text{min},b})$$

Pair score

$$D = d / \sqrt{(\sigma_{\text{master}}^2 + \sigma_b^2)} < 3$$

Updated master position

$$x_{\text{master}} = \Sigma(x_b / \sigma_b^2) / \Sigma(1 / \sigma_b^2)$$

High-precision positions get more weight.

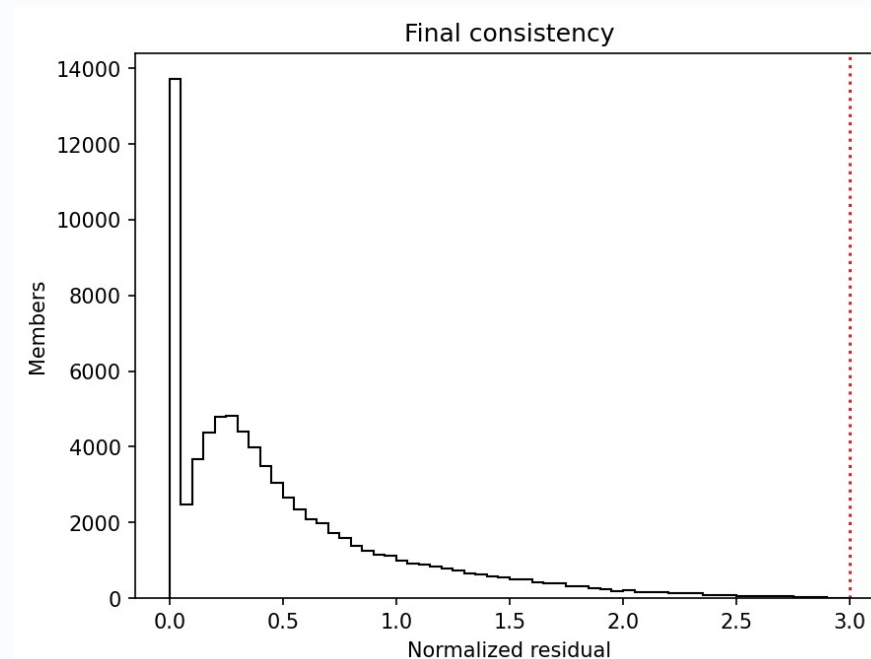
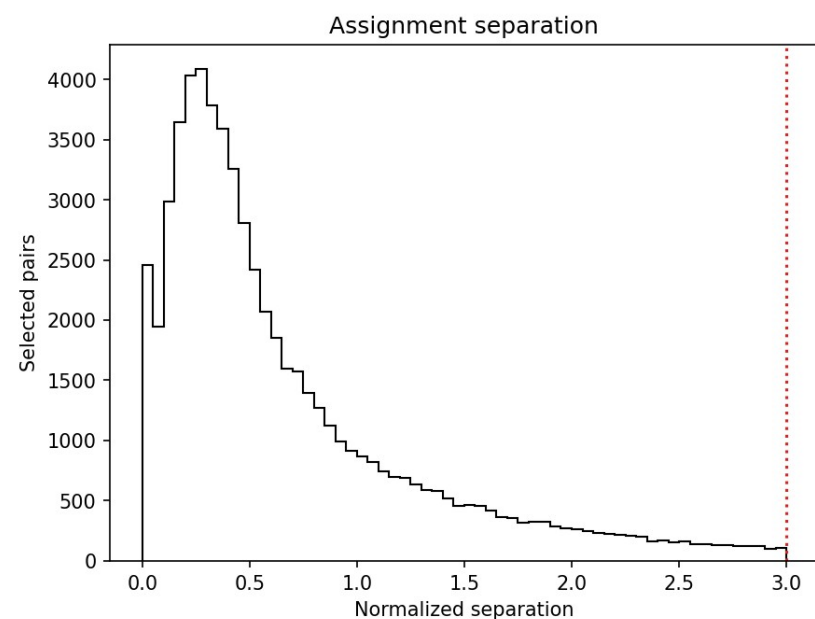
7DT Matching Validation

Accepted members agree with their final master position

99.85%

within 3σ of the final master

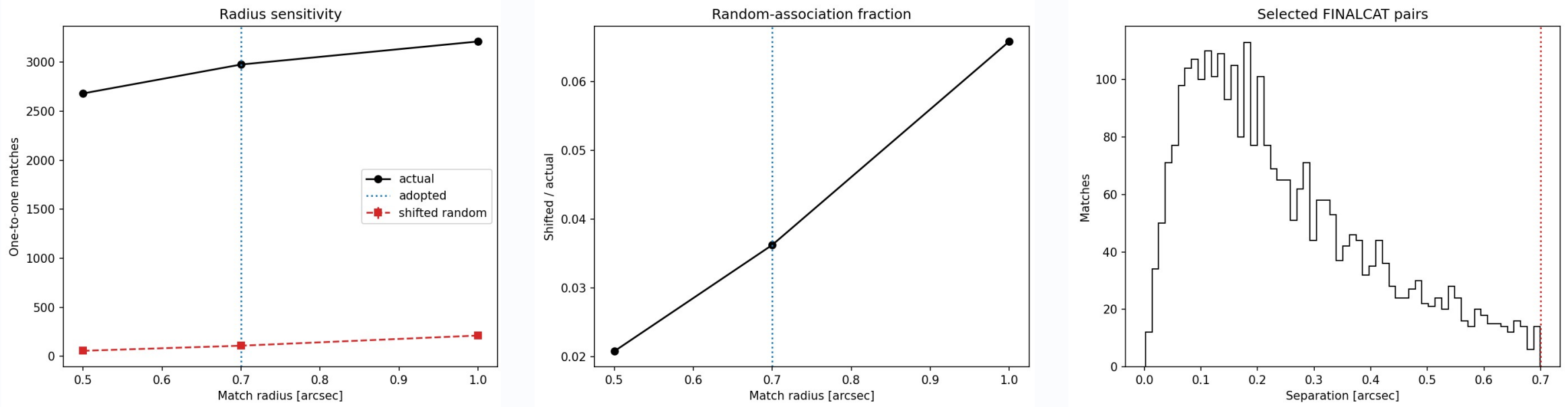
117 suspect members of 78,882 detections



19,489 masters · 1,157 detected in all 20 bands · $R = d_{\text{final}} / \sqrt{(\sigma_{\text{final}}^2 + \sigma_b^2)}$

FINALCAT Match Radius

The random-shift test supports a 0.7" sky-match radius

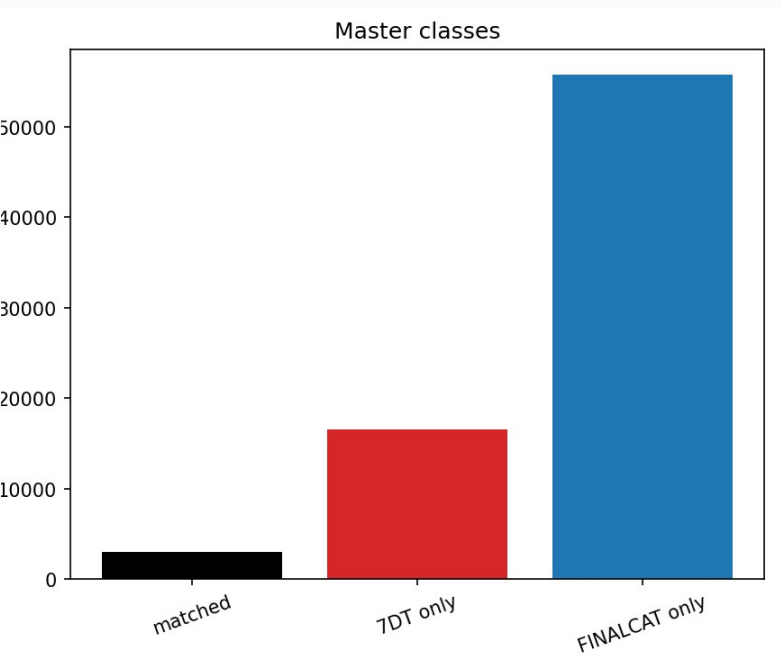
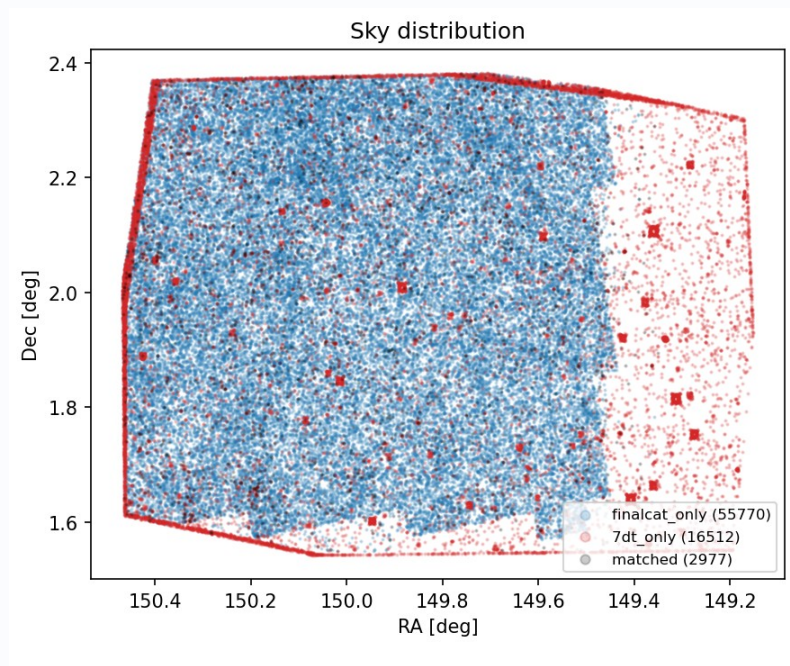


Radius	Actual matches	Shifted random	Random / actual
0.5"	2,682	55.8 ± 5.2	2.08%
0.7"	2,977	107.9 ± 7.2	3.62%
1.0"	3,212	211.4 ± 16.9	6.58%

Eight 10" or 20" shifts estimate chance associations.

Final Union Catalog

2,977 shared objects in a 75,259-object catalog



Master class	Objects
7DT + FINALCAT	2,977
7DT only	16,512
FINALCAT only	55,770
Total union catalog	75,259